

PARTE 2

Condense los grados de libertad utilizando MAPLE 12 u otro programa

*Maple 12 - C:\Users\Ivan\Desktop\Segundo Evaluación 201630811.mw - [Server 1]

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Units (SI)

Units (FPS)

Common Symbols

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> with (LinearAlgebra):
> Kbb := Matrix([[4*h^2, 0, 2*h^2, 0], [0, 6*h^2, h^2, 0], [2*h^2, h^2, 8*h^2, h^2], [0, 0, h^2, 6*h^2]]);

```

$$K_{bb} = \begin{bmatrix} 4h^2 & 0 & 2h^2 & 0 \\ 0 & 6h^2 & h^2 & 0 \\ 2h^2 & h^2 & 8h^2 & h^2 \\ 0 & 0 & h^2 & 6h^2 \end{bmatrix} \quad (1)$$

```

> Kbb_inv := LinearAlgebra:-MatrixInverse(Kbb);

```

$$K_{bb_inv} = \begin{bmatrix} \frac{23}{80h^2} & \frac{1}{80h^2} & -\frac{3}{40h^2} & \frac{1}{80h^2} \\ \frac{1}{80h^2} & \frac{41}{240h^2} & -\frac{1}{40h^2} & \frac{1}{240h^2} \\ -\frac{3}{40h^2} & -\frac{1}{40h^2} & \frac{3}{20h^2} & -\frac{1}{40h^2} \\ \frac{1}{80h^2} & \frac{1}{240h^2} & -\frac{1}{40h^2} & \frac{41}{240h^2} \end{bmatrix} \quad (2)$$

```

> Kba := Vector([y[1], y[2], y[3], y[4]]);

```

$$K_{ba} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (3)$$

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$$Kbb_{inv} := \begin{bmatrix} -\frac{3}{40 h^2} & -\frac{1}{40 h^2} & \frac{3}{20 h^2} & -\frac{1}{40 h^2} \\ \frac{1}{80 h^2} & \frac{1}{240 h^2} & -\frac{1}{40 h^2} & \frac{41}{240 h^2} \end{bmatrix}$$

> $Kba := \text{Vector}([y[1], y[2], y[3], y[4]]);$

$$Kba := \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (3)$$

> $Kba := \text{Vector}([6 * h, 6 * h, 6 * h, 6 * h]);$

$$Kba := \begin{bmatrix} 6 h \\ 6 h \\ 6 h \\ 6 h \end{bmatrix} \quad (4)$$

> $Rl := Kbb_{inv} \cdot Kba;$

$$Rl := \begin{bmatrix} \frac{57}{40 h} \\ \frac{39}{40 h} \\ \frac{3}{20 h} \\ \frac{39}{40 h} \end{bmatrix} \quad (5)$$

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Units (SI)
 $[\text{unit}]$ $[m]$ $[s]$
 $[N]$ $[kg]$ $[Pa]$
 $[W]$ $[J]$ $[K]$
 $[T]$ $[A]$ $[V]$
 $[C]$ $[\Omega]$ $[F]$
 $[H]$ $[rad]$ $[sr]$
 $[mol]$ $[lx]$ $[im]$
 $[S]$ $[Wb]$ $[Np]$

Units (FPS)
 $[\text{unit}]$ $[ft]$ $[s]$
 $[poundal]$ $[lb]$
 $\left[\frac{\text{poundforce}}{\text{inch}^2}\right]$ $[HP]$
 $[poundal \cdot ft]$ $[rad]$
 $[sr]$

Common Symbols
 π e i j I ∞
 Σ Π $\int$ d $\cap$ $\cup$
 $\geq$ $>$ $\neq$ $\leq$ $<$
 $\nless$ $\nless$ α $\approx$ $\sim$ $=$
 $\neq$ $\equiv$ $\neq$ $\in$ $\notin$ $\subseteq$

$R1 := Kbb_inv.Kba;$

$$R1 := \begin{bmatrix} 6h \\ \frac{57}{40h} \\ \frac{39}{40h} \\ \frac{3}{20h} \\ \frac{39}{40h} \end{bmatrix} \quad (5)$$

$Kabx := \text{Vector[row]}([x[1], x[2], x[3], x[4]]);$

$$Kabx := \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \quad (6)$$

$Kab := \text{Matrix}([6*h, 6*h, 6*h, 6*h]);$

$$Kab := \begin{bmatrix} 6h & 6h & 6h & 6h \end{bmatrix} \quad (7)$$

$R2 := Kab.R1;$

$$R2 := \frac{297}{20} \quad (8)$$